

Practice Lab:

DSE Troubleshooting, Tuning

- This Practice Lab is dependent on Discussion Unit 6292, but also on most other Discussion Units we have presented.
- In this Practice Lab, you are given an (n) node DSE cluster in need of attention.
- You are done when:
 - All servers are running
 - Metrics are being collected
 - Loads are balanced between nodes
 - You can answer the questions that follow
 - List all issues you found, and what/how you had to do to fix

The Challenge:

Part 1

- You will be given an (n) node cluster
- Ops Center will be running on one node with a public IP
- You don't have to "fix" Ops Ctr
- *Any problems are with the non Ops Ctr nodes*

Data (or similar) you will receive-

```
-----BEGIN RSA PRIVATE KEY-----  
MIIEpAIBAAKC ... deqw8hD8BB1HJV++HDG3DwL1SyR  
...  
9Iduj4WNK4yZ ... WvIE4yLa/OZZuG2WYRucUhQQ==  
-----END RSA PRIVATE KEY-----
```

```
df, 9 instances  
tw, 9 instances  
cluster name: xyz  
instance type: n1-standard-2  
Node 0:  
  public hostname: 35.193.183.32  
  public ip: 35.193.183.32  
  private hostname: tony ... les.internal  
  private ip: 10.240.0.24  
  region: Google  
Node 1:  
  public hostname: 35.193.152.40  
  public ip: 35.193.152.40  
...
```

The Challenge: Part 1, Repair the cluster

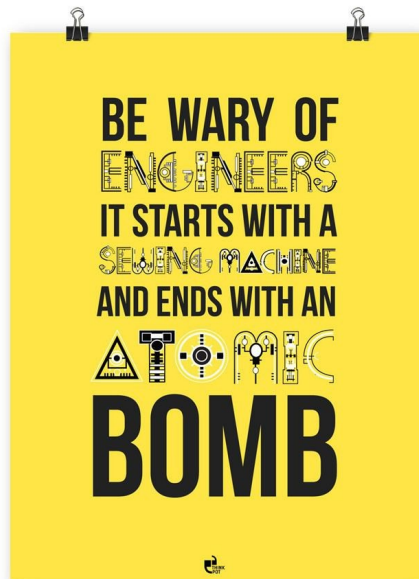


- Don't just blow the cluster away.
- Look first. Many persons create more problems initially through incorrect assumptions. Know what and why you are changing something.
- Use nodetool heavily; also CQLSH
- If you know exactly what to do, 20 minutes tops.
- Most persons take 2-4 hours for part 1.

The Challenge: Part 1, Share, but be wary ..

- Share with other attendees, certainly
- The clusters **were** all identical.

Then your neighbor did something wrong, and will share details with everyone of incorrect assumptions/steps. Dave !



The Challenge: Part 2, Tuning/cassandra-stress

- Monitor with Ops Center (should be running on node0 on the public ip):
 - Reads, Average reads, read timeouts
 - Writes, Average writes, write timeout
 - Disk utilization, Disk IO
 - CPU Utilization
 - Java Memory, Java Garbage collection
 - Other Metrics that you like
- Run cassandra-stress with a replication factor of 2 to each data center
 - Load at least an average of 1 GB of data per node (Approximately 10,000,000 records with default stress table)
 - Use skills gained from cassandra-stress exercise

The Challenge: Part 2, Be Prepared to Report



Hermes Greek God of Commerce, Communications and Wealth
Source: <https://www.majesticdragonfly.com/hermes-greek-god-statue-8220>

- What is the fastest read speed you can get out of the system at 90%, 95%, max ?
- What is the fastest write speed you can get out of the system at 90%, 95%, max ?
- What is the average data size difference between nodes, smallest dataset, largest, the difference between the two ?
- Can you make that difference closer or even exact ? If so how ?
- Is it worth the effort to do so ?

The Challenge: Part 2, Be Prepared to Report

- What problems did you find?
- What was the hardest part to fix?



Source: <https://www.popculturecrossing.com/classics/thepaperchase1973review>

Practice Lab:

Lessons Learned



